Time series: ARMA process

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The homework is provided on the last slide

ARMA process

A stochastic process x_t, $t \in \mathbb{Z}$, is called an **autoregressive moving average** process, ARMA(p, q), with zero mean if x_t is stationary in the wide sense and if for every t,

$$x_t - a_1 x_{t-1} - a_2 x_{t-2} - \ldots - a_p x_{t-p} = \epsilon_t + b_1 \epsilon_{t-1} + \ldots + b_q \epsilon_{t-q},$$

where ϵ_t is white noise with mean 0 and variance σ^2, $a_p \neq 0$, $b_q \neq 0$ and polynomials

$$a(z) = 1 - a_1 z - a_2 z^2 - \ldots - a_p z^p$$

and

$$b(z) = 1 + b_1 z + \ldots + b_q z^q$$

have no common factors (roots).

$p, q \in \{0, 1, 2, 3, \ldots\}$.

R script of simulating an ARMA process

Task. Plot a realization of the ARMA(3,2) process

$$x_t - 0.4x_{t-1} + 0.2x_{t-2} - 0.1x_{t-3} = \epsilon_t + 0.6\epsilon_{t-1} - 0.3\epsilon_{t-2}$$

for $t = 4, 5, \ldots, 100$ and $x_1 = 1$, $x_2 = -2$, $x_3 = 0.5$; $\sigma = 1.2$.

```
® n=100
eps=rnorm(n, 0, 1.2)
x=rep(0,n)
x[1]=1; x[2]=-2; x[3]=0.5;
for(t in (4:n))
  x[t]=0.4*x[t-1]-0.2*x[t-2]+0.1*x[t-3]+
       eps[t]+0.6*eps[t-1]-0.3*eps[t-2];
plot((1:n), x, pch=19, main="ARMA(3,2) process
       a(1,-0.4,0.2,-0.1) b(1,0.6,-0.3)")
lines((1:n), x)
```

ARMA process and backward shift operator

The operator B (defined for elements of a time series) is called the **backward shift operator** if

$$Bx_t = x_{t-1}.$$

Note that $B^0 x_t = x_t$ and

$$B^j x_t = \underbrace{B(B(\ldots B}_{j \text{ times}}(x_t)\ldots)) = x_{t-j}.$$

Thus, the ARMA equation

$$x_t - a_1 x_{t-1} - a_2 x_{t-2} - \ldots - a_p x_{t-p} = \epsilon_t + b_1 \epsilon_{t-1} + \ldots + b_q \epsilon_{t-q},$$

can be rewritten in the form

$$a(B)x_t = b(B)\epsilon_t,$$

$$(1 - a_1 B - a_2 B^2 - \ldots - a_p B^p)x_t = (1 + b_1 B + \ldots + b_q B^q)\epsilon_t.$$

ARMA process with mean μ and particular cases

A stochastic process x_t is called an ARMA(p, q) **process with mean** μ if the process $y_t = x_t - \mu$ is an ARMA(p, q) process with zero mean.

A stochastic process x_t is called an **autoregressive process**, AR(p), if x_t is an ARMA$(p, 0)$ process, i.e. $a(B)x_t = \epsilon_t$.

A stochastic process x_t is called an **moving average process**, MA(q), if x_t is an ARMA$(0, q)$ process, i.e. $x_t = b(B)\epsilon_t$.

Existence and uniqueness

From definition of an ARMA process, it follows that a process satisfying

$$x_t - a_1 x_{t-1} - a_2 x_{t-2} - \ldots - a_p x_{t-p} = \epsilon_t + b_1 \epsilon_{t-1} + \ldots + b_q \epsilon_{t-q},$$

for all t, may be stationary or non-stationary, depending on coefficients.

Theorem. A stationary solution of the ARMA equations exists (and is also the unique stationary solution) if and only if

$$a(z) = 1 - a_1 z - a_2 z^2 - \ldots - a_p z^p, \quad z \in \mathbb{C},$$

has no roots on the unit circle of $\mathbb{C}$, that is, $a(z) \neq 0$ for all $|z| \neq 1$.

Then ARMA x_t can be written in the form $x_t = \sum_{j=-\infty}^{\infty} c_j \epsilon_{t-j}$ because $1/a(z)$ can be expanded in a (Taylor) series.

Causality

$$x_t - a_1 x_{t-1} - a_2 x_{t-2} - \ldots - a_p x_{t-p} = \epsilon_t + b_1 \epsilon_{t-1} + \ldots + b_q \epsilon_{t-q},$$

An ARMA(p, q) process x_t is called **causal**, if there exist constants $c_0, c_1, c_2, \ldots$ such that $\sum_{j=0}^{\infty} |c_j| < \infty$ and

$$x_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j}$$

for all $t \in \mathbb{Z}$.

$$\ldots, x_{-2}, x_{-1}, x_0, x_1, x_2, \ldots, x_{t-1}, \underbrace{x_t}_{\|}, x_{t+1}, \ldots$$

$$\underbrace{\ldots, \epsilon_{-2}, \epsilon_{-1}, \epsilon_0, \epsilon_1, \epsilon_2, \ldots, \epsilon_{t-1}, \epsilon_t}, \epsilon_{t+1}, \ldots$$

Lemma

Causality is equivalent to the condition of absence of roots of $a(z) = 1 - a_1 z - a_2 z^2 - \ldots - a_p z^p$ inside the unit circle of $\mathbb{C}$, that is, $a(z) \neq 0$ for all $|z| \leq 1$.

Invertibility

$$x_t - a_1 x_{t-1} - a_2 x_{t-2} - \ldots - a_p x_{t-p} = \epsilon_t + b_1 \epsilon_{t-1} + \ldots + b_q \epsilon_{t-q},$$

An ARMA(p, q) process x_t is called **invertible**, if there exist constants $s_0, s_1, s_2, \ldots$ such that $\sum_{j=0}^{\infty} |s_j| < \infty$ and

$$\epsilon_t = \sum_{j=0}^{\infty} s_j x_{t-j}$$

for all $t \in \mathbb{Z}$.

$$\ldots, \epsilon_{-2}, \epsilon_{-1}, \ \epsilon_0, \ \epsilon_1, \epsilon_2, \ldots, \epsilon_{t-1}, \underbrace{\epsilon_t}_{\|}, \epsilon_{t+1}, \ldots$$

$$\underbrace{\ldots, x_{-2}, x_{-1}, x_0, x_1, x_2, \ldots, x_{t-1}, \ x_t}, \ x_{t+1}, \ldots$$

Lemma

Invertibility is equivalent to the condition of absence of roots of $b(z) = 1 + b_1 z + \ldots + b_q z^q$ inside the unit circle of $\mathbb{C}$, that is, $b(z) \neq 0$ for all $|z| \leq 1$.

Spectral density

Theorem

The spectral density of ARMA(p, q) process x_t satisfying

$$x_t - a_1 x_{t-1} - a_2 x_{t-2} - \ldots - a_p x_{t-p} = \epsilon_t + b_1 \epsilon_{t-1} + \ldots + b_q \epsilon_{t-q},$$

has the form

$$f_x(\lambda) = \frac{\sigma^2 |b(e^{-i\lambda})|^2}{2\pi |a(e^{-i\lambda})|^2},$$

where i is the imaginary unit and ϵ_t is a white noise with zero mean and variance σ^2.

For an MA(1) process $x_t = \epsilon_t + b_1 \epsilon_{t-1}$, the spectral density is

$$f(\lambda) = \frac{\sigma^2}{2\pi} |1 + b_1 e^{-i\lambda}|^2 = \frac{\sigma^2}{2\pi} (1 + 2b_1 \cos(\lambda) + b_1^2).$$

For an AR(1) process $x_t - a_1 x_{t-1} = \epsilon_t$, the spectral density is

$$f(\lambda) = \frac{\sigma^2}{2\pi |1 - a_1 e^{-i\lambda}|^2} = \frac{\sigma^2}{2\pi (1 - 2a_1 \cos(\lambda) + a_1^2)}.$$

Covariance function

A causal ARMA(p, q) process x_t can be represented as

$$x_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j}.$$

Theorem. The covariance function of a causal ARMA process is

$$R(k) = \sigma^2 \sum_{j=0}^{\infty} c_j c_{j+|k|}.$$

Theorem. The covariance function of a MA(q) process $x_t = b(B)\epsilon_t$ is

$$R(k) = \begin{cases} \sigma^2 \sum_{j=0}^{q-|k|} b_j b_{j+|k|} & |k| \leq q, \\ 0 & |k| > q. \end{cases}$$

See an example in Lecture notes 4.

Two opposite problems

Problem 1

Given: Coefficients $a_1, \ldots, a_p$ of an AR(p) process.
Task: Find the covariance function $R(k)$.
Solution: see the top of slide 10.

Problem 2

Given: The order p of an AR(p) process and the covariance function $R(k)$.
Task: Find the coefficients $a_1, \ldots, a_p$.
Solution: Solve the Yule-Walker equations.

The Yule-Walker equations can also be used for Problem 1.

Problem 2 is needed for estimating $a_1, \ldots, a_p$ on the base of $\hat{R}(0), \hat{R}(1), \ldots, \hat{R}(k)$.

Yule-Walker equations

Multiplying both sides of the AR(p) equation

$$x_t = a_1 x_{t-1} + \ldots + a_p x_{t-p} + \epsilon_t$$

by x_{t-k} and taking the expectation, we get

$$\mathbf{E} x_{t-k} x_t = a_1 \mathbf{E} x_{t-k} x_{t-1} + \ldots + a_p \mathbf{E} x_{t-k} x_{t-p} + \mathbf{E} x_{t-k} \epsilon_t.$$

Since $\mathbf{E} x_{t-k} \epsilon_t = \begin{cases} \sigma^2, & k=0 \\ 0, & k>0 \end{cases}$ and $\mathbf{E} x_s x_t = R(t-s)$, we have

$$R(k) = a_1 R(k-1) + a_2 R(k-2) + \ldots + a_p R(k-p), \quad k > 0$$

and $R(0) = a_1 R(1) + a_2 R(2) + \ldots + a_p R(p) + \sigma^2, \ k = 0.$

The system of the **Yule-Walker equations** is

$$\begin{cases} R(1) = a_1 R(0) + a_2 R(1) + \ldots + a_p R(p-1) \\ R(2) = a_1 R(1) + a_2 R(0) + \ldots + a_p R(p-2) \\ \vdots \\ R(p) = a_1 R(p-1) + a_2 R(p-2) + \ldots + a_p R(p-p) \end{cases}$$

Dividing everything on $R(0)$, we obtain equations with $\rho(k)$.

Homework with AI

Use AI tools to deepen your understanding of the ARMA models.

- **Causality and Invertibility:** Provide a specific ARMA(2,1) model to an AI (e.g., $x_t - 0.5x_{t-1} + 0.6x_{t-2} = \epsilon_t + 0.4\epsilon_{t-1}$) and ask it to determine if the model is causal and invertible by finding the roots of the characteristic polynomials.
- **Yule-Walker Derivation:** Ask the AI to show the step-by-step derivation of the Yule-Walker equations for an AR(2) process.
- **Simulation:** Ask the AI to generate a script that slowly changes an AR(1) coefficient from 0.1 to 0.9 and plot the resulting time series to visualize how "memory" increases.
- **Spectral Analysis:** Ask the AI to compare the spectral density of an AR(1) process with a positive coefficient versus one with a negative coefficient. Request the R-code to plot both and explain why the peaks appear at frequency 0 vs frequency 0.5.